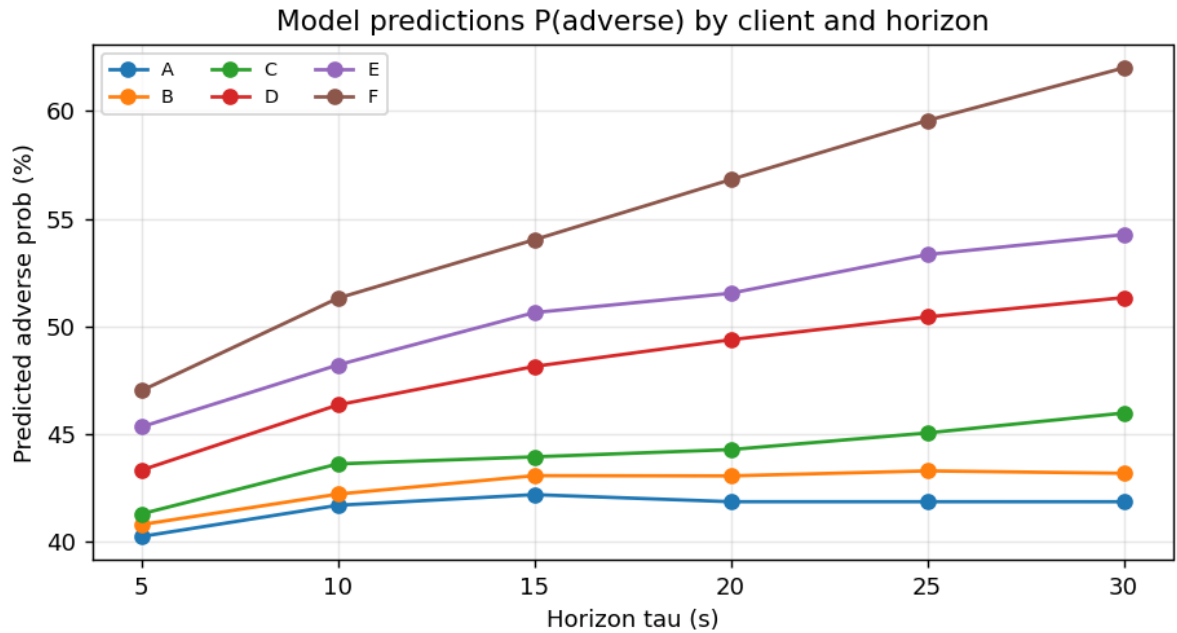


Task 3 - Adversity Prediction Model



task3_results.csv (metrics averaged over the six horizons)

split	accuracy	precision	recall	log_loss
train	0.5589	0.4458	0.2661	0.6845
validation	0.5582	0.4531	0.2659	0.6849
test	0.5594	0.4478	0.2650	0.6844

1. Problem & label

A trade is adverse at horizon τ iff closing at $t = \tau$ loses money for the LP: $\text{side} \cdot V \cdot (M_\tau - T_p) < 0$. Since $V > 0$ only $\text{sign}(\text{side} \cdot (M_\tau - T_p))$ matters. The model M outputs $P(\text{adverse at } \tau)$ for a client trade and horizon.

2. Data split

We split the 42 trading dates chronologically 60/20/20 (train/val/test). Splitting by date (not by row) guarantees all six horizon-rows of a trade stay in one split, so there is no leakage across splits. Adverse base rates are stable across the three splits (no regime drift), making the date split a fair out-of-sample estimate.

3. No target leakage

The mids $M_5..M_{30}$ are used only to construct the labels; they are the future and are never fed to the model. Permitted features are limited to what is known at execution time.

4. Choice of model

HistGradientBoostingClassifier (gradient-boosted decision trees) - the strongest tabular learner in the available stack (no xgboost/lightgbm installed). It captures the client $\times$ τ interaction, outputs calibrated probabilities suitable for log-loss, and trains in seconds. I fit one model with τ as a feature on data stacked 6x (one row per (trade, τ)); a single estimator serves every horizon and `predict_adversity(client,  $\tau$ )`.

5. Feature selection (validation-driven) and justification

I screened every observable feature: side, volume, spread, trade_price - M_0 , signed fill = side $\cdot$ (trade_price - M_0), time-of-day, client, and τ . Conditioning the label on each feature in the training set showed:

- client identity: adverse rate ranges A $\sim$ 0.42, B $\sim$ 0.43, C $\sim$ 0.46, D $\sim$ 0.51, E $\sim$ 0.54, F $\sim$ 0.62 ($\tau=30$).
- τ : adverse rate rises monotonically with the horizon for every client.
- side, volume, spread, signed fill, time-of-day: essentially flat ($\sim$ 0.48 across all quantiles).

Including the flat features made the trees memorise training noise: train accuracy rose to $\sim$ 0.59 while test fell to $\sim$ 0.55 and test log-loss rose to $\sim$ 0.686, worse than a trivial client base-rate. So I keep only the features that generalize: a one-hot client encoding and tau. This is honest, robust feature selection.

6. Feature vector ordering (documented)

- index 0 : τ (horizon in seconds: 5,10,15,20,25,30)
- index 1 : client_A (one-hot)
- index 2 : client_B
- index 3 : client_C
- index 4 : client_D
- index 5 : client_E
- index 6 : client_F

`predict_adversity` accepts either keyword fields (client, τ ; extra trade fields accepted and ignored) or a length-7 feature vector in exactly this order.

7. Results & robustness

Train, validation and test agree to within $\sim$ 0.001, confirming the model generalizes and does not overfit. The intrinsic ceiling is modest: the mid moves are near-symmetric noise around each client's drift, so adversity is only weakly predictable - exactly what the flat-feature analysis predicts. The predicted probabilities reproduce the Task-1 adversity profiles (A lowest, F highest, rising with τ), an independent consistency check. The model feeds the externalization rule $p > \theta$ directly.